

# Assessment and Formation under Agentic AI: Risks, Possibilities, and a Certification Architecture for Student-Centred Higher Education

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## Abstract

Agentic frontier AI systems reached general availability in June 2026. These systems can carry out long-horizon autonomous work, and they can simulate the *process* of intellectual labour — drafts, revisions, reasoning traces — as readily as its products. This paper analyses what that means for students in higher education, and proposes an institutional response.

Two theses organize the analysis. The first concerns validity. Assessment evidence fails to the degree that its production is unsupervised and its medium is one a frontier model can generate, and the failure is governed by how *common* machine production becomes, not by the mere fact that it is possible. A likelihood-ratio model makes this precise. Before AI, wholesale outsourcing of assessed work ran at roughly 2 percent; the share of UK students reporting that they directly include AI-generated text in assessed work has risen from 3 to 8 to 12 percent across the 2024–2026 survey waves. Indicators of this kind set bounds on how common the invalidating behaviour is; none of them measures it exactly. But the model shows what rising prevalence does, whatever its exact level: unsupervised artifacts drift toward carrying no information about the student at all. Two corollaries follow. The first is that polished work loses its evidential value before anything else does. The second is that, under plausible assumptions about who adopts these tools, suspicion of excellent work becomes statistically rational — and its burden falls on students whose unaided work happens to resemble machine output, a group that includes the genuinely excellent and non-native writers of English. The second thesis concerns formation. Entry-level cognitive work has always financed the development of expertise: employers paid for novice labour, and novices built capability by doing it. To the extent that agentic systems strip the market value from novice output, that financing fails. The work of forming experts then migrates — by market logic, not by anyone's policy choice — to institutions that charge for formation. Universities are the incumbent such institutions. Organizing the resulting risks around the goods students stand to lose yields a further prediction: the only assessment that survives the validity collapse is human-intensive, and human-intensive means expensive. Unless secured assessment is funded as an entitlement, the *validity* of a credential threatens to become a positional good — something available where it can be afforded.

The constructive contribution is a certification architecture built for this situation. It states three registers of competence as distinct validity arguments; specifies *delegation judgment* — the quality of a student's decisions about delegating work to machines, verifying what comes back, and owning the result — as a certifiable construct, measured by a fault-seeded audit instrument scored with signal-detection methods and anchored in live defence; and certifies it through *second-order entrustment*, in which the institution licenses the student as the entruster

of machine work. A graduated developmental sequence and explicit data-governance conditions complete the design. Falsifiers are stated for each major claim.

*Keywords:* agentic AI, assessment validity, delegation judgment, appropriate reliance, entrustment, higher education, cognitive offloading, academic integrity

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## 1. Introduction

In June 2026, Anthropic released Claude Fable 5, the first publicly available model of its Mythos class, alongside a more capable variant restricted to vetted organizations (Anthropic, 2026a). The release is best understood not as a sudden discontinuity — the vendor’s own system card places the models on the capability trendline observed across their predecessors (Anthropic, 2026b) — but as the moment a particular class of system arrived at consumer scale. This paper calls that class *agentic frontier systems*. Five of their properties matter for education. They sustain long autonomous runs of work rather than answering single questions. They generate process as readily as product: drafts, false starts, and revision histories are simply more output. They persist across sessions, so the system a student consults in week ten can know everything from week one. By vendor report — not yet independently replicated — they have begun to participate in question formation, the stage of inquiry education has treated as definitionally human. And they arrive inside explicit governance tiers, which ration capability by organization, subscription, and even query domain. Each of these properties has educational consequences on its own. Their conjunction is what is new.

Most institutional analysis of AI in higher education proceeds within one of three framings, and each centres something other than the student. An *integrity framing* centres the institution’s certification function, and treats students primarily as suspects. A *productivity framing* centres efficiency, and treats students as future workers to be tooled up. A *literacy framing* centres the technology itself, and treats students as operators to be trained. Each framing captures something real. None begins from the question a student-centred analysis has to begin from: what goods do students seek and deserve from higher education, and how does this class of system change their ability to obtain them? Biesta (2009) offers a serviceable map of those goods: qualification (the development and certification of capability), socialization (induction into communities and practices), and subjectification (the formation of independent persons). The argument of this paper is that agentic systems destabilize all three at once — qualification through the collapse of assessment evidence (Section 3), socialization through the erosion of the apprenticeship structures embedded in entry-level work (Section 4), and subjectification through pressure on the epistemic and relational conditions under which independent judgment forms (Section 5).

Beyond its synthesis, the paper makes four contributions. First, it formalizes the assessment problem. It states the two conditions under which evidence fails — unsupervised production, in a machine-generable medium — and embeds them in a likelihood-ratio model of how evidential value decays as machine production spreads. The model’s comparative statics turn out to explain several otherwise puzzling pathologies of the current enforcement regime (Section 3; Appendix B). Second, it derives, rather than asserts, the migration of formation toward universities: the

migration follows from human-capital theory once the market value of novice output collapses, and the mechanism organizes the otherwise conflicting labour-market evidence (Section 4). Third, it identifies an equity risk that follows from the paper's own analysis. If valid assessment is necessarily human-intensive, then validity acquires a price, and a price implies a distribution; the paper names this the *validity divide* and states the prediction in falsifiable form (Sections 5.4 and 8). Fourth, and centrally, it specifies *delegation judgment* — the quality of a student's decisions about what cognitive work to delegate to machines, how to verify what comes back, and how to own the integrated result — as a certifiable construct. The construct is defined against its nearest neighbours in three literatures, theorized as metacognition extended beyond the brain's boundary, measured by a specified instrument (Appendix A), and certified through a decision rule this paper calls *second-order entrustment*, which inverts the direction of the entrustment frameworks now emerging in professional education (Section 7). Sections 6 and 8 complete the analysis with possibilities that go beyond the conventional menu of AI-in-education applications, and with a research agenda whose falsifiers are stated.

## 2. Related Work

**Assessment validity under generative AI.** The position that assessment should be designed to secure valid inferences, rather than to police misconduct, is by now well established. Dawson (2021) distinguished assessment security from academic integrity before the generative wave arrived. Dawson, Bearman, Dollinger, and Boud (2024) state the position directly: assessments that depend on students not using AI, but cannot prevent that use, “are not particularly useful for high-stakes assessment of learning.” Corbin, Dawson, and Liu (2025) document why instruction-based control fails and argue that only structural changes to tasks bind behaviour. In concurrent independent work, Weidlich (2026) articulates AI-era assessment threats within Kane's (2013) argument-based validity framework, cataloguing threats inference by inference. At the level of institutional design, the University of Sydney's two-lane model (Liu & Bridgeman, 2023; Bridgeman, Liu, & Weeks, 2024) and the regulator-commissioned Australian guidance (Lodge et al., 2023, 2025) operationalize programme-level security, and England's quality body has converged on similar advice while explicitly deprecating detection software (QAA, 2023). The present paper inherits this position throughout. What this paper adds is the thing an enumerated catalogue does not contain: a single mechanism that generates the threats. The mechanism is the failure of one warrant — attribution — under two stated conditions, and once it is stated, the rate of failure and its comparative statics follow from it. The field's own research agenda asks for exactly this kind of account: work that identifies “where validity breaks down” in hybrid human–AI production, with shared language for evaluating evidence quality there (Corbin et al., 2025).

**Judgment constructs in assessment.** Evaluative judgement — the capability to make decisions about the quality of one's own and others' work — is a mature construct in higher-education research (Tai, Ajjawi, Boud, Dawson, & Panadero, 2018), and it has been explicitly extended to generative AI, including judgements about AI-use processes and about whether and when use is appropriate (Bearman, Tai, Dawson, Boud, & Ajjawi, 2024). Walton, Bearman, Crawford, Tai, and Boud (2025) document empirically the consequential, largely unguided

judgement events students experience when working with these systems. On the task side, the Artificial Intelligence Assessment Scale assigns permitted-use levels to assessment tasks (Perkins, Furze, Roe, & MacVaugh, 2024; Perkins, Roe, & Furze, 2025). Medical education contributes the certification machinery this paper builds on: entrustment, in which certifying a trainee means documenting a willingness to trust them with a defined professional activity at a defined supervision level (ten Cate, 2005, 2013), embedded in programmatic assessment, whose proportionality principle ties the stakes of a decision to the volume of evidence beneath it (van der Vleuten et al., 2012). Entrustment has recently been extended to AI — but with the tool as the trustee whose use must be safeguarded (Gin et al., 2025), or with proficient technology use treated as one coarse entrustable activity (Jacobs et al., 2023). What none of these frameworks provides is the inverse decision higher education now needs: certifying the student as the *delegator* of machine work, on a defined construct, with a specified evidence model. Building that is Section 7's task.

**Reliance on automated advice.** Human–automation research supplies forty years of warnings that the present moment is confirming. Automation converts operators into monitors while removing the practice that monitoring presupposes (Bainbridge, 1983). Trust must be calibrated to a system's actual capability for reliance on it to be appropriate (Lee & See, 2004; Parasuraman & Riley, 1997). The recent human–AI interaction literature has made appropriate reliance measurable — accepting correct advice one would have gotten wrong, rejecting incorrect advice one would have gotten right (Schemmer et al., 2023) — and its findings are sobering in a consistent direction. Showing people explanations makes them accept advice more across the board; it does not help them tell good advice from bad (Bansal et al., 2021). Interventions reduce overreliance without making reliance more appropriate (Bo, Wan, & Anderson, 2025). People delegate poorly by default, largely because they lack accurate models of their own capabilities (Fügener, Grahl, Gupta, & Ketter, 2022). Behavioural evidence at scale now exists for the failure mode this paper centres: in preregistered experiments where an assistant's accuracy was covertly randomized, untrained adults followed confidently wrong machine output on roughly four of five trials in which they consulted it, and access to the assistant inflated their confidence rather than calibrating it (Shaw & Nave, 2026 — a preprint, with a published critique in Shimabukuro, 2026). Reliance measurement has reached student populations (Pitts, Rani, Mildort, & Cook, 2025; Hou et al., 2025), and evidence-centered design has been applied to human–AI collaborative writing (Cheng, Lyons, Chen, Gašević, & Swiecki, 2024). What no prior work does, on searches conducted in June 2026, is connect behavioural reliance measurement to the certification decisions assessment must actually make. That connection is this paper's central construction.

**Labour-market context.** Payroll evidence documents concentrated employment declines for early-career workers in AI-exposed occupations (Brynjolfsson, Chandar, & Chen, 2025). Danish registers linked to adoption surveys find near-null average effects and challenge the causal attribution at the entry margin (Humlum & Vestergaard, 2026); aggregate occupational-mix indicators remain flat (The Budget Lab at Yale, 2026). Field research documents what happens to trainees when technology severs the expert–novice bond: excluded from legitimate practice, they

resort to the norm-violating workarounds Beane calls shadow learning (Beane, 2019, 2024). Section 4 supplies the financing mechanism that organizes these findings, and states the evidence that would decide between the competing readings.

### 3. The Collapse of Distant Proxies

Section 1 claimed that agentic systems destabilize qualification through the collapse of assessment evidence. This section makes that case in four steps. Section 3.1 locates the warrant on which assessment inference depends. Section 3.2 states the two conditions under which that warrant fails. Section 3.3 models how fast the failure progresses. Section 3.4 identifies what survives, and what the survival costs.

#### 3.1 Assessment as inference and the attribution warrant

Assessment never measures learning directly. It measures observable performances and infers unobservable constructs from them, and validity is a property of those inferences and their uses, not of instruments (Messick, 1995, p. 741). Two features of Messick’s unified account matter for this paper. The account includes the consequences of score use inside the concept of validity itself — a feature Section 5.4 relies on. And it treats validity as “an evolving property”, which means that the same instrument, unchanged, can lose validity when the world changes around it. That second feature is the door this section walks through.

Kane’s (2013) argument-based framework supplies the working machinery. On Kane’s account, a score interpretation is a chain of inferences — scoring, generalization, extrapolation, decision — and each link rests on warrants that can be separately interrogated. The framework’s governing principle is blunt: “a chain of reasoning is only as strong as its weakest link” (p. 64).

The warrant that agentic systems break is one so basic it is rarely stated. Call it the *attribution premise*: the assumption that the cognitive decisions constituting a submitted performance were the student’s own. The premise was never certain; contract cheating and ghostwriting predate AI by centuries. It was, however, probabilistically adequate, and for a reason: circumvention was costly, risky, and unevenly available, and so its base rate stayed low. What follows formalizes how that adequacy fails, and how fast.

#### 3.2 Two conditions

A natural first statement of the failure would say that evidence fails when it is evaluated asynchronously in a machine-generable medium. That statement founders on an instructive counterexample. The invigilated examination is graded asynchronously, in prose, and plainly survives. What protects it is not when it is graded but how it is *produced*. The operative conditions are these:

*An evidentiary proxy loses its connection to individual competence to the degree that (a) its production is unsupervised and (b) it is expressible in a medium a frontier model can generate. Once systems meeting (b) are cheap and ubiquitous, the evidential value of every*

*proxy meeting both conditions decays with the prevalence of machine production — toward zero.*

Call proxies meeting both conditions *distant proxies* — distant in supervision, not in time. The take-home essay, the problem set, and the project report are all distant proxies, and that they are failing is close to consensus (Dawson et al., 2024; Corbin, Dawson, & Liu, 2025; Weidlich, 2026).

The structural point that distinguishes this collapse from every previous integrity problem concerns *process* evidence. The dominant institutional response of 2023–2025 relocated evidential weight from products to processes — drafts, version histories, reflective commentaries. The theory behind that move was that simulating weeks of intellectual struggle costs more than undergoing it. Agentic systems invert the economics. A system performing long-horizon autonomous work generates the intermediate states as a matter of course, because intermediate states are simply more output. Process artifacts are therefore distant proxies too. They fall together with the products they were meant to authenticate, and they fall by one mechanism.

The collapse bears a family resemblance to Goodhart-type measure corruption (Strathern, 1997, p. 308). The resemblance is real, but the dynamics differ. Grades degraded historically through gaming and expectation inflation, which is a slow social process. Distant proxies degrade through simulation of the evidence itself, and that proceeds at the speed of tool adoption.

### 3.3 Prevalence, not possibility

The principle stated in Section 3.2 is a limit result. It says what happens to evidential value as machine production becomes universal; it does not say what is true today. Limit results invite a fair objection: simulation was always possible, yet essays carried information for decades. The objection identifies exactly the right quantity. What governs evidential value is not whether machine production is possible but how common it has become.

Appendix B states the model that makes this precise; its logic can be given briefly in plain language. An artifact is evidence about competence through its likelihood ratio — through how much more probable a given level of quality is when the work comes from a competent student than when it comes from a less competent one. In other words, an artifact tells the assessor something only insofar as its quality depends on who produced it. Unaided production links quality to competence, so the ratio does evidential work. Machine production severs the link. For undirected end-to-end simulation, the quality of the output is a property of the model and the task, not of the student. Appendix B.1 states this *severance* premise and scopes it carefully.

What an assessor actually observes is a mixture of the two channels. Some share of submitted artifacts is produced unaided and some share is simulated, so the observed evidence blends the two in proportion to their prevalence. Call the simulation share  $\sigma$ . The model's first result says, in plain terms, that evidence quality falls as simulation spreads and vanishes in the limit. Formally, diagnosticity is a decreasing function of  $\sigma$ , converging to nothing as  $\sigma$  grows (Result 1). The more simulated work enters the pool, the less any artifact can say about the student who submitted it.

The second result says the decay is not uniform: polished work loses its evidential value first. The reason is mechanical. Frontier models produce polished work, so the high-quality region of

the distribution is where the two channels pool first and completely — where a flawless artifact stops telling the assessor which channel produced it. Flawed, idiosyncratic, characteristically human work retains evidential value longest (Result 2). Excellence stops discriminating before difficulty does.

The third result depends on who adopts. Suppose adoption concentrates where its private return is largest — plausibly among students for whom the capability gap is largest. That pattern is consistent with experimental findings that AI effects concentrate along prior-capability lines (Bastani et al., 2025; Lehmann, Cornelius, & Sting, 2025). If the supposition holds, polished work becomes evidence *against* unaided competence before it becomes uninformative (Result 3). On that adoption pattern, polish starts to count against the student before it stops counting at all.

Result 3 explains a familiar pathology. The enforcement-era heuristic of suspecting work that is “too good” is, on this analysis, rational inference under a collapsing mixture. But its victims are systematic: they are the students whose unaided work resembles model output. That class includes the genuinely excellent. And because detectors are biased against low-perplexity prose — prose a language model finds statistically predictable — it also includes non-native writers whose competence is real (Liang et al., 2023). Validity failure and procedural injustice are projections of the same mixture.

How fast is  $\sigma$  moving? The pre-AI baseline is well measured. Across 65 studies and 54,514 students, self-reported commercial contract cheating averaged 3.52 percent (Newton, 2018). In post-2014 samples the figure was 15.7 percent, and rising. That synthesis aggregates heterogeneous question wordings, however, and for wholesale outsourcing the right anchor is the per-artifact measure. On that anchor, the largest single survey found 2.2 percent of students reporting having obtained an assignment to submit (Bretag et al., 2019). Roughly two-thirds of those students reported submitting it. The pre-AI base rate, in short, sat near 2 percent, and three brakes held it there: price, exposure risk, and an unambiguous norm.

Each brake is now released. The price brake is gone, because marginal cost approaches zero inside subscriptions students already hold. The exposure-risk brake is gone too, because detection has failed in the field. When entirely GPT-4-written submissions were injected into a real university examination system, 94 percent went undetected — and they outscored real students (Scarfe, Watcham, Clarke, & Roesch, 2024). Where detectors are deployed, they are trivially evaded and demographically biased (Liang et al., 2023; Weber-Wulff et al., 2023). Watermarking, honestly summarized, is safe on the false-positive side, but it is not universal and it is not strongly robust (Kirchenbauer et al., 2024; Zhang et al., 2024; Dathathri et al., 2024). The norm, finally, has dissolved from a boundary into a gradient: 94 percent of UK undergraduates use generative AI in some way when preparing assessed work (Stephenson & Armstrong, 2026).

Honesty about measurement is required at this point, because no current statistic measures  $\sigma$  as the model defines it. To make the claim measurable, this paper uses a three-way taxonomy throughout, defined by the *locus of cognitive decisions* — by where the decisions that constitute the work actually sit. In *assistance*, the constitutive decisions stay with the student. In *co-production*, they are distributed between student and machine. In *simulation*, they transfer to the machine end-to-end. In the model’s terms,  $\sigma$  is the prevalence of the third category. The available

indicators do not estimate  $\sigma$ . What they do is bound the *invalidating channels* — the forms of use that break the attribution premise.

Three such indicators are on the record. The first comes from students' own reports. Self-reported direct inclusion of AI-generated text in assessed work has risen across the 2024–2026 UK survey waves, from 3 to 8 to 12 percent. Fifteen percent of the same population consider it acceptable (Stephenson & Armstrong, 2026, pp. 13–14). The second comes from detector-side telemetry, which shows submissions with more than 80 percent AI-written content rising from 3.3 to roughly 15 percent (Turnitin, 2026). Citing a detector statistic may seem inconsistent, since this section has just argued that detectors cannot be trusted. The inconsistency is only apparent, and the reason is worth a sentence. A detector too unreliable to accuse any individual student can still track a population trend, because the trend question tolerates errors the accusation question cannot: evasion means the detector undercounts, false positives mean it overcounts, and neither bias plausibly invents a fivefold rise. The third indicator comes from platform usage. Before agentic tooling matured, a quarter of higher-education conversations on one frontier platform were already direct output creation (Anthropic, 2025a).

The counterweight is also on the record. US secondary surveys found overall cheating prevalence roughly flat across the ChatGPT transition (Lee, Pope, Miles, & Zárate, 2024). Wholesale substitution remains minority behaviour everywhere it has been measured.

The claim, therefore, is not that most students simulate. It has three parts. The first is that artifact validity depends on  $\sigma$ . The second is that every structural force on  $\sigma$  points up, while every historical brake is gone. The third is that the observed indicator levels already matter quantitatively. The invalidating channels stand at twelve to fifteen percent on the available indicators. That is six times the pre-AI base rate, reached within three years. If those indicator levels are read as the simulation share — a reading the indicators only bound, not establish — then under Appendix B.6's worked parameterization the diagnosticity of polished distant artifacts would already have roughly halved. If that reading held, in other words, roughly half the evidential value would already be gone. On the observed trajectory, the decay continues toward zero. For high-stakes decisions resting on a single artifact, the burden of proof has shifted to anyone still relying on them. Section 8 states the falsifier, its measurement design, and its threshold.

### 3.4 What survives, and what it costs

What survives the collapse is what fails condition (a): production that is supervised. The viva, the invigilated examination, the supervised demonstration, the studio critique, and the clinical observation all keep their evidential value, because a human witnesses the production. These formats should be interactive where the construct demands probing depth and transfer (Joughin, 1998), and not otherwise. Sector practice is converging on the same conclusion. The Australian regulator-commissioned guidance builds programme-level security at decision points, and its three implementation pathways all depend on secure tasks assuring learning outcomes (Lodge et al., 2023, 2025). England's quality body recommends synoptic assessment and deprecates



detection (QAA, 2023). Practical guidance on oral examination at scale now exists (Kalman & Wright, 2026; Barba & Stegner, 2026).

Three concessions accompany the survival claim. The first is that the surviving instruments have known reliability and bias problems of their own. Those problems are addressed by structure, training, recording, and appeal — not by romanticism (Joughin, 1998). The second is that supervised performances are still proxies. Supervision restores the attribution warrant; it does not make competence directly observable. The third concession is the costly one. Distant proxies were assessment's cheap evidential technology, and supervised performance requires synchronized human attention — someone must be present while the work is produced. When the only valid assessment is costly assessment, validity itself becomes a scarce good with a price and a distribution. Section 5.4 develops that equity consequence, and Section 7.1 prices it.

## 4. The Conditional Ladder

Section 3 concerned how institutions certify competence. This section concerns how competence is formed in the first place — and, above all, who has paid for forming it. Public commentary on AI and entry-level work has converged on an image for what is at stake: a career ladder whose bottom rungs are being removed, leaving novices no way to climb toward expertise. This paper adopts the ladder as its framing device and makes the image precise. Made precise, the bottom rungs are paid entry-level cognitive work. Employers paid for novice labour, and novices built capability by doing it. The argument of this section runs in four steps. To the extent that agentic systems strip the market value from novice output, the financing of those rungs fails (4.1). The labour-market evidence bearing on this prediction must be read on both sides (4.2). Supervising machines does not become a new bottom rung (4.3). And the assignment higher education inherits is conditional, distributional, and reflexive (4.4).

### 4.1 Who paid for formation

Novices have always built expertise by performing genuine but low-stakes versions of expert work under supervision. The structure has been formalized repeatedly: as the apprenticeship sequence of modelling, coaching, and fading (Collins, Brown, & Newman, 1989); as legitimate peripheral participation (Lave & Wenger, 1991); as development within the zone of proximal development (Vygotsky, 1978). Beneath all three formulations sits an economic precondition that is easy to miss. Someone must pay for novice work.

Human-capital theory says who paid, and how. The skills built by entry-level cognitive work are *general*: they raise a worker's value to many employers, not just to the firm where the training happens. Competitive firms do not pay for general training. Becker (1962) stated the result plainly: "the cost as well as the return from general training would be borne by trainees, not by firms." Trainees paid by accepting wages below their current productivity. Formation, in other words, was financed out of the value of the novice's own output. Imperfect labour markets offer a second channel. There, firms can finance general training and recoup the cost through post-training wage compression — by paying trained workers less than their enhanced productivity —

an arrangement that requires retention (Acemoglu & Pischke, 1998, 1999). Even in that channel, the trainee must “take part in some productive activities” for the wage-discount mechanism to operate.

Both channels narrow under the same shock. If agentic systems drive the marginal product of novice cognitive work toward zero — or below the cost of supervising it — then Becker’s wage discount must go negative. A trainee whose output is worth nothing would, in effect, have to pay the firm to be trained. Negative training wages are blocked by wage floors, by credit constraints, and by labour law. The firm-financed channel narrows too, because it is weakest precisely in high-mobility cognitive labour markets. The conclusion follows to the extent that the auxiliary premises hold, and they deserve explicit statement: the skills are general; the value of novice output collapses; wage floors bind; rent recapture is limited. Granted those premises, formation migrates out of employment and into institutions where people pay to undergo supervised formation on work of little market value.

The market is already growing such institutions wherever universities have left room — bootcamps, premium apprenticeships, income-share arrangements. Their growth confirms the mechanism rather than refuting it, because each is tuition-financed formation under another name. Universities inherit the ladder not because they are best suited to it, but because they are the incumbent institutions whose business model survives novice work having no market value.

#### 4.2 The evidence, on both sides

The mechanism just stated is a prediction about labour markets, and the evidence bearing on it genuinely points in more than one direction. This subsection sets it out on both sides.

On the side of the thesis, payroll records covering millions of US workers show employment for early-career workers — those aged 22 to 25 — in the most AI-exposed occupations declining roughly 6 percent from late 2022 through September 2025 (Brynjolfsson, Chandar, & Chen, 2025). Older workers in the same occupations grew 6 to 9 percent over the same window. Under firm-time controls, the gap amounts to a relative decline of about 16 percent. In the raw series, young software developers fell nearly 20 percent from the late-2022 peak. The declines concentrate where AI usage is automative rather than augmentative. One interpretation the authors offer is the one this thesis predicts: that early-career knowledge is disproportionately the codified knowledge models replace. They canvass alternatives, however, and they concede that under the strictest controls, significance arrives only after 2024 (Brynjolfsson, Chandar, & Chen, 2026).

The countervailing evidence is substantial, and it concerns the right margins to different degrees. Danish registers linked to adoption surveys find precise near-null average effects of chatbot adoption on earnings and hours through late 2024 (Humlum & Vestergaard, 2026). The same study replicates the US early-career decline in exposed occupations — and then shows that the decline is *not concentrated in adopting firms*. Aggregate US occupational-mix indicators remain flat, and the compositional shifts that do exist predate generative AI (The Budget Lab at Yale, 2026). And at the macroeconomic level, the estimated ceilings on aggregate effects are modest (Acemoglu, 2025).

The stock-flow distinction reconciles part of this conflict. Incumbent workers are the stock; new entrants are the flow; null effects on incumbents are exactly what a mechanism operating on the entry flow predicts. The distinction does not cover the Danish non-concentration finding, however, which is entry-margin evidence. The mechanism's answer is that hiring contraction operates through forward-looking hiring plans and market-level adjustment, rather than through adopting firms' own headcount. That answer is a real general-equilibrium defence. It is also the kind of defence that could insulate the theory from any firm-level test, so it pays rent in advance. The rent takes the form of two commitments. If firm-linked designs using direct agentic-tooling telemetry also show no entry-margin concentration as adoption matures, that counts against the mechanism. And Section 8 adds a price-margin falsifier — relative entry wages — that hiring-plan stories cannot dodge.

One absence in the evidence deserves separate notice. The relocated ladder has not appeared anywhere: nothing yet documents entry-level roles systematically rebuilt as paid apprenticeships in agent supervision. The next subsection argues that the absence is predictable rather than temporary.

### **4.3 Why supervising the machine is not a new bottom rung**

The optimistic reading holds that juniors will supervise agent fleets, and that supervision will become the new apprenticeship. The objection to that reading is forty years old, and it comes with a mechanism. Automation removes precisely the routine engagement through which operators maintain the skills that supervision presupposes. The monitor is asked to judge decisions she has been prevented from learning to make — in Bainbridge's words, "the human monitor has been given an impossible task" (Bainbridge, 1983, p. 776). Her own example of unexercised knowledge decaying is, fittingly, "any subject which you passed an examination in at school and have not thought about since" (p. 775).

The cognitive version of the objection is sharper. Monitoring requires a task model. Task models are built by doing the task. Delegation removes the doing.

The empirical signature exists at both ends. At one end, trainees are structurally excluded from practice when technology concentrates execution in expert hands, and they learn instead through norm-violating workarounds (Beane, 2019, 2024). At the other end, untrained adults followed confidently wrong machine output on four of five engaged trials when they consulted it (Shaw & Nave, 2026). That second finding comes from a preprint, the tasks were short, and the setup was an engineered worst case; this paper carries those caveats deliberately.

Employers who wished to rebuild apprenticeship inside AI-saturated workflows would therefore have to deliberately preserve novice practice that the technology makes economically pointless. That is exactly the general-training cost Becker's logic says competitive firms will not bear. The relocated ladder is not impossible. It is unfunded.

### **4.4 The educational consequence: conditional, distributional, reflexive**

Three properties of the inherited assignment matter for design, and they give this subsection its title.

The assignment is conditional. Where regulation mandates supervised practice — in licensed professions and clinical placements — the market ladder persists. The thesis accordingly predicts formation migrating fastest exactly where payroll declines concentrate, and that gradient is testable.

The assignment is distributional. The withdrawn subsidy was real money, and its withdrawal shifts the costs of expertise reproduction onto tuition payers and public funders. It does so at the same moment Section 3 forces assessment onto expensive channels. Who pays is therefore a first-order question, not a footnote; Sections 5.4 and 7.1 take it up.

And the assignment is reflexive. Teaching assistantships, sessional contracts, and postdoctoral positions are the academy's own entry-level cognitive work. The same capabilities that strip the subsidy from junior analysts strip it from the future examiner supply on which any assessment architecture depends. Section 7.4 attaches a governance instrument to that exposure.

One reframing completes the analysis. A student who delegates everything is a rational actor responding to a payoff matrix in which the credential still pays, the practice no longer visibly does, and the labour market is broadcasting that the entry-level work is going anyway. Signalling theory implies that this equilibrium degrades the credential for everyone (Spence, 1973; Appendix B.7). A student-centred institution treats that as a design failure to be engineered out, by changing what the payoff matrix rewards, rather than as a character failure to be policed. Section 7 is the engineering.

## 5. Risks to Student Goods

Risk registers in this domain are usually organized by institutional exposure. This section organizes them differently: by the student good at stake. Each risk is graded by the strength of the evidence behind it, as *established*, *emerging*, or *theoretical*.

### 5.1 Epistemic goods

The core epistemic risk is not the obvious one. The danger is not that students will believe false outputs. It is that the activity of evaluating will be displaced — unevenly, and invisibly — by the activity of accepting.

The strongest behavioural evidence for this displacement comes from three preregistered experiments with a GPT-4o assistant (Shaw & Nave, 2026). The design detail that matters is the randomization. The assistant's accuracy was covertly randomized item by item, so its reliability on any given item could not be predicted from the items before it. Whatever discrimination participants showed therefore had to come from evaluating the answer in front of them. The experiments enrolled 1,372 participants, who completed 9,593 trials between them.

What participants did with the help is the finding. On the trials where the assistant was accurate, participants who consulted it followed it 92.7 percent of the time. On the trials where it was confidently wrong, they followed it 79.8 percent of the time. Accurate or not, the assistant was followed far more often than not.

Confidence moved the same way. Access to the assistant raised participants' confidence by nearly twelve points, despite half of the answers being faulty, and confidence did not decline as the faulty answers accumulated. Access inflated confidence; it did not calibrate it.

Pooling the trials on which participants engaged a wrong assistant gives the distribution of outcomes. In 73.2 percent of those trials, participants showed cognitive surrender — they adopted the wrong answer. In 19.7 percent they showed cognitive offloading, overriding the assistant correctly. The remaining 7.1 percent were failed overrides. Surrender was not the exception; it was the norm.

Three caveats hold this finding at the *emerging* grade. The study is a preprint. The tasks were short reasoning items. And the 50-percent error rate was engineered, far above field conditions (see also Shimabukuro, 2026).

What matters for assessment design survives those caveats. When participants received per-item incentives together with immediate feedback, their overrides of faulty output more than doubled, rising from 20.0 to 42.3 percent. A forty-four-point accuracy gap remained even so. The lesson is that discrimination responds to accountability structures but is not produced by them alone.

Convergent evidence of lower strength points in the same direction. Lee et al. (2025) studied 319 knowledge workers, who reported on 936 of their own real tasks. Higher confidence in the AI was associated with less self-reported enacted critical thinking, with effort shifting toward verification and oversight. Gerlich (2025) finds that frequent AI use correlates with lower critical-thinking scores, partially mediated by offloading. The same study finds the youngest cohort reporting both the heaviest use and the lowest scores. Its design is self-report, and it supports no causal inference. And a small EEG study found weaker connectivity and impaired ownership among LLM users (Kosmyna et al., 2025). That study had 54 participants, of whom 18 were in the critical crossover session. Its result is a hypothesis-generating signal, and it is under methodological challenge (Stanković, Hirche, Kollatzsch, & Doetsch, 2025).

Two further risks now have direct evidence, and they share a mechanism. The first is homogenization. Generative assistance raises the quality of individual work while reducing the diversity of the collective output (Doshi & Hauser, 2024), and the diversity loss is localized to feedback-tuned models (Padmakumar & He, 2024). The second is calibration corruption: people absorb external answers into their sense of their own knowledge (Ward, 2021; cf. Sparrow, Liu, & Wegner, 2011, whose offloading experiments stand even though the paper's first experiment failed replication).

The shared mechanism is preference training. Reward models favour responses that match user beliefs over truthful ones (Sharma et al., 2024). The same training signal that flatters a student's weak idea also narrows the distribution of ideas across students. And systems optimized for persuasion become less accurate as they become more persuasive (Hackenburg et al., 2025).

None of this requires the model to be wrong. The deepest statement of the risk concerns systems that are fluent, helpful, and genuinely accurate. Such systems present the characteristics that human epistemic vigilance reads as trust signals, without those characteristics carrying their usual information — what Maynard (2026) calls “honest non-signals.” This is a hypothesis-grade

mechanism. If it were borne out, it would explain why sophisticated users may be least protected, and why verification is harder to teach than the literacy framing assumes.

The section's grading can now be stated. Sycophancy and homogenization count as established in experimental settings. The displacement of evaluation counts as emerging. The vigilance bypass remains theoretical.

## 5.2 Developmental goods

The strongest causal evidence on developmental harm is a field experiment conducted in Turkey, whose PNAS publication is the version of record (Bastani et al., 2025). Relative to students who never had access, students given unrestricted GPT-4 access during practice performed 48 percent better while assisted. Their subsequent unassisted performance was 17 percent lower. The same students looked stronger with the tool and were weaker without it. A guardrailed tutor variant raised practice performance by 127 percent and eliminated the harm.

The mechanism is the central dissociation of the learning sciences: performance during practice and learning from practice come apart, and conditions that inflate the former routinely fail to produce the latter (Soderstrom & Bjork, 2015). Complementary experiments locate the harm and the benefit in how the system is used (Lehmann, Cornelius, & Sting, 2025). The harm sits in solution-seeking and copy-paste use. The benefit sits in explanation-seeking use. The same experiments find prior-knowledge gaps widening. Those findings are exploratory analyses, run within experiments whose preregistered effect of access on overall learning was null. The signal, in other words, lives in the usage pattern, not in access.

The equity implication is uncomfortable. Assistance compresses *visible* performance while leaving the underlying distribution alone, or widening it. The evidence for the compression is exploratory — dispersion analyses, of assisted practice only (Bastani et al., 2025). The result is a Matthew dynamic for cognitive capital (Stanovich, 1986, extending Merton, 1968). Students with strong self-regulation convert these systems into amplifiers. Students without it delegate the load-bearing struggle and cannot verify what comes back.

The formation-window harm is established in the studied context; the distributional dynamic is emerging. The design conclusion follows from the pattern. The documented harm is not assistance as such; it is *uninstrumented substitution during formation windows*. Design separates the outcomes.

## 5.3 Relational goods

Always-available systems occupy the default-confidant position by structural advantage. They impose zero scheduling cost, they carry zero judgment risk, and their patience is infinite. The evidence on what follows from this is genuinely mixed, and this section grades it accordingly, as emerging with mixed valence.

On the displacement side, computing undergraduates re-route their help-seeking away from peers and instructors and toward the model, and they report isolation (Hou et al., 2025). Among US teens, 72 percent have used AI companions. A third have used them for social interaction and

relationships, and “they don’t judge me” is among the reasons they give (Common Sense Media, 2025).

On the benefit side, AI companions reduce loneliness in the short run (De Freitas et al., 2026). And US students’ chatbot help-seeking fell from 30 to 17 percent over a single year. Even now, 84 percent still turn first to people (Tyton Partners, 2025).

Whatever the balance turns out to be, three losses remain the design targets. The first is the loss of mentorship externalities: the supervisor who never sees the confused draft cannot spot the struggling student, and cannot write the informed reference. The second is the loss of formative friction, because preference-trained systems structurally under-supply the pushback that peers and teachers provide with stakes attached (Sharma et al., 2024). The third is belonging. The student-centred question is therefore not whether AI confidants are bad. It is whether institutions track what their availability does to the human relational infrastructure that students are also paying for.

#### 5.4 Positional goods and the validity divide

Credentials are substantially positional (Spence, 1973), and three pressures now converge on them. From below comes signal erosion (Section 3). From above comes labour-market contraction (Section 4). And across the student body itself runs capability stratification: students do not reach these systems on equal terms. Access to these tools tracks household social grade. Institutional provision reaches 38 percent of students. And safeguard routing falls back to older models at roughly twice the vendor-reported average rate on exactly the technical tasks students bring (Stephenson & Armstrong, 2026; Artificial Analysis, 2026). Even among students who have access, the capability actually delivered differs.

To these three pressures this paper adds a fourth, which follows from its own analysis: **the validity divide**. A note on the name is owed at first use. The term is this paper’s coinage, and the echo of the digital-divide family of concepts is deliberate. Three established neighbours should be held apart from it. It is not the digital divide in access to digitally based assessment. It is not psychometrics’ “differential validity,” which names differential predictive accuracy across demographic groups. And it is not the familiar gap between how researchers and practitioners understand validity. What the term names is the stratification derived below: a divide in who can afford assessment that remains valid.

The derivation runs from the paper’s own results, and it is short. Section 3.4 established that the assessment which survives is human-intensive and priced. Whatever is priced is distributed. The equilibrium to fear is therefore not the absence of valid certification but its stratification. In that equilibrium, human-examined formation concentrates in wealthy institutions, while mass and distance provision certifies through cheap machine-mediated channels — with surveillance as the budget substitute for supervision. Assessment validity becomes a positional good. The question is not only what assessment can know, but who can afford to be validly known.

One qualification belongs here. Stratification of assessment intensity by institutional wealth predates AI. That is why Section 8 states the prediction as a change from a baseline, not as the appearance of something new.

Signalling theory sharpens the prediction. Where evaluators sort candidates directly, the candidates who are hardest to evaluate are the ones left bearing the residual signalling burdens. In hiring models this is a worker-level result (Kurlat & Scheuer, 2021, §6); this paper extends it interpretively, to certification access for distance students, part-time students, and under-resourced institutions. The prediction is falsifiable, and Section 8 states it in that form. The design answer — certification floors as funded entitlements — is built into Section 7.1.

### 5.5 Procedural goods

The post-2023 enforcement regime measurably damaged due process. The damage begins with the detectors. They are systematically biased: their average false-positive rate on essays by non-native English writers is 61.3 percent (Liang et al., 2023). The corresponding rate for native writers is 5.1 percent. They are also unreliable as standalone evidence (Weber-Wulff et al., 2023). Institutions that act on them anyway convert a statistical artefact into accusation at scale. The UK recorded roughly 7,000 proven cases in 2023–24 (The Guardian, 2025). That is about 5.1 per thousand students. Field tests, meanwhile, imply that the caught fraction is small (Scarfe et al., 2024).

Chilling effects are visible on both sides of the Atlantic. Among UK students, fear of accusation is the most-cited deterrent to legitimate use, and it is reported even by non-users (Stephenson & Armstrong, 2026). In the US, 43 percent of students report not using generative AI at all under widespread prohibition regimes. The surveyors attribute that figure partly to misconduct fear suppressing honest reporting (Muscanell & Gay, 2025).

Appendix B.4 gives the structural reading. Under a collapsing mixture — that is, as machine production spreads through the pool of submitted work — suspicion of polished work is rational inference. And the incidence of that suspicion is unjust by construction. Surveillance responses double the harm without restoring the evidence. Proctoring imposes disparate burdens by disability, caregiving, and bandwidth (Swauger, 2020), and it fails against simulation.

The durable exit is to stop staking high-stakes decisions on unverifiable claims. That means redesigning what is certified, which is the work of Section 7. It also means collecting any instrumented evidence under the firewall conditions of Section 7.2, so that the apparatus that certifies delegation judgment cannot become the proctoring regime this section indicts. Detector bias and chilling effects are graded as established.

## 6. Possibilities Beyond the Conventional Menu

The risks of Section 5 are not the whole account; this section turns to what the same systems make possible. Institutional adoption so far has drawn on a conventional menu of applications: chatbot tutoring, automated feedback, and productivity tooling. That menu under-uses the capability class this paper is about — the agentic frontier systems described in Section 1, which sustain long autonomous runs of work, generate process as readily as product, and persist across sessions. Four families of possibility go further. Each carries a named failure mode, and where a possibility depends on which capability tier an institution can reach, that conditionality is flagged.



**Compressed research apprenticeships.** If hypothesis generation, literature synthesis, and analysis pipelines are available at marginal cost, the delay between entering a discipline and *doing* it can compress. A second-year student can run a full inquiry cycle that previously required doctoral infrastructure. Course-based undergraduate research experiences supply the chassis (Auchincloss et al., 2014). What agentic tooling attacks is the cost structure that kept them scarce.

This is the strongest constructive answer to Section 4. If the economy no longer supplies novice rungs on the ladder of Section 4, the curriculum can supply participant rungs higher up.

The capability-tier conditionality bites here. Genuinely novel hypothesis generation is an announced capability of the gated tier only, and the possibility is therefore contingent on institutional access agreements.

The failure mode to guard against is verification debt. The debt accrues when students orchestrate inquiry beyond their evaluative competence and launder model errors into confident conclusions. That warning has a lineage in the automation literature (Bainbridge, 1983; Parasuraman & Riley, 1997). The design implication — verification capacity certified before orchestration privileges expand — is Section 7.3's.

**Instrumented apprenticeship at scale.** The tutoring literature, restated at its true size, supports structure over species: how the tutoring is organized matters more than what kind of tutor delivers it. The numbers behind that claim are worth setting out one at a time. Human tutoring delivers roughly 0.79 standard deviations over no tutoring, on ten comparisons. Step-based intelligent tutoring systems deliver roughly 0.76, on twenty-eight comparisons. Answer-based systems deliver 0.31, on one hundred and sixty-five (VanLehn, 2011). Modern PreK–12 field RCTs pool at 0.37 (Nickow, Oreopoulos, & Quan, 2024). And Bloom's (1984) two-sigma result rested partly on a mastery-threshold confound that VanLehn judges sufficient to account for tutoring's advantage over classroom mastery learning. Read together, the comparisons support the opening claim: the structure is doing the work.

Structure of this kind is what frontier models can deliver at scale — if it is deliberately built in. The 2025 evidence is consistent with that reading. A purpose-built AI tutor outperformed active-learning classes across two single-lesson units, at 0.63 standard deviations with the estimate ceiling-suppressed (Kestin et al., 2025). Teacher-supervised sessions with a general-purpose model produced roughly 0.31 standard deviations in Nigerian schools, at exceptional cost-effectiveness (De Simone et al., 2025). And the guardrailed arm of the Turkish trial removed the harm its raw arm produced.

Two caveats temper the picture. Neither arm of the Turkish trial out-learned the no-AI controls. And no trial yet shows instrumented access *teaching* where raw access erodes — in one design, in higher education, at retention distance. Section 8 registers that missing experiment.

The ordering across these studies converges even so: instrumented access lands at or above the no-AI condition, and raw access lands below it. The load-bearing institutional act is therefore the provisioning of pedagogically governed configurations, not the provisioning of access.

The failure mode to guard against here is the default-interface problem. No instrumented configuration survives offering a worse user experience than the raw default — unless assessment

values what the configuration builds.

**Evaluative pedagogy: the student as examiner.** The third family positions students as critics and auditors of machine output: find the flaw in this proof, break this analysis, audit this pipeline.

Stated honestly, the warrant is suggestive. The protégé effect is an effort finding, with partial learning effects, in middle-school samples (Chase, Chin, Oppezzo, & Schwartz, 2009). And the ICAP framework predicts the deepest engagement only for genuinely *dialogic* designs (Chi & Wylie, 2014). In ICAP's terms, solo critique is constructive, not interactive. A computer agent counts as a partner only when its responses are content-relevant turns.

The design implication is exact. Build dialogic audit loops, in which the student challenges, the model defends, and the student presses — not solo error-hunts. The deepest engagement mode is also the mode that generates the behavioural evidence Register C needs (Section 7.2).

The failure mode to guard against here is staleness: a critique target that never changes soon stops exercising critique. The remedy is to regenerate critique targets per student and per occasion, which agentic systems make cheap.

**The studio university and capability stewardship.** Hours spent on functions frontier systems perform adequately — content transmission, routine feedback, administrative triage — can be reallocated to functions no model performs. Those functions are relational: standing in accountable relation to students, vouching for them, convening the communities where professional identity forms. The reallocation is also evidential, because the studio, the lab meeting, and the viva are where Section 3's surviving assessment lives.

Early telemetry warns that the drift runs the wrong way by default. Educators automate grading at among the highest rates of any task, while rating AI least suited to it (Anthropic, 2025b). Section 7.4 therefore attaches a reporting instrument.

The same institutional position grounds stewardship. Universities are the class of organization that vendors and governments gate advanced capability toward, and that position can be exercised on students' behalf. Provisioned access can neutralize the subscription divide. Half of students think institutions should provide the tools; 38 percent receive them (Stephenson & Armstrong, 2026). Pedagogically governed configurations and audit-log co-ownership can be set as procurement conditions, alongside non-training clauses and student seats in governance.

The hard case is distance and part-time provision, where synchronous assessment is costliest and surveillance temptations are strongest. There, stewardship either proves itself or exposes the architecture as a residential privilege. Section 7.1's certification floor is addressed to exactly this case.

Two failure modes attend this family, and both are captures. In the first, cost-cutters capture the reallocation. In the second, the vendor captures the partnership. Both are met structurally, by multi-vendor postures, public configuration policies, and the explicitly legitimate option of *designed non-use*.

## 7. A Certification Architecture

The response this analysis demands operates at the level of what is certified, not at the level of how suspicion is policed. The architecture proposed here has four components, and they interlock. Section 7.1 states what is certified, and under what validity arguments. Section 7.2 specifies the construct that makes augmented work certifiable, together with the instrument that measures it. Section 7.3 supplies the developmental sequencing that connects formation to certification. Section 7.4 states the governance that keeps the whole apparatus on the right side of its own principles.

### 7.1 Three registers of competence, as three validity arguments

The starting point is a design the field already has. The two-lane model, which separates secured assessment from open assessment (Liu & Bridgeman, 2023), correctly identified the surviving evidential channel. Its critics are also correct: a binary division invites students to pass through the open lane without engagement, and it erases the pedagogical middle ground (Curtis, 2025). The resolution proposed here is that the open lane conflates two certifiable constructs, and that the two should be separated. Each register is therefore stated as its own interpretation/use argument in Kane's (2013) sense, with its own evidence conditions, its own decision rule, and its own threats. Stated that way, the critics' objections stop being objections. They convert into design requirements.

**Register A — unaided capacity.** *Claim:* The student can perform the discipline's non-delegable core without machine participation. *Evidence conditions:* Production is supervised, in the sense of Section 3.4, and it is interactive wherever the construct demands probing. Examiners are trained, sessions are recorded, and decisions are appealable. *Decision rule:* Gates at programme decision points are conjunctive — each must be passed in its own right — but every gate rests on programmatically aggregated, multi-occasion evidence. This follows the proportionality principle of programmatic assessment: “only low-stake decisions can be based on single data points, whereas all high-stake decisions require input from many” (van der Vleuten et al., 2012, p. 207).

Both halves of that decision rule carry weight, and each guards against a different failure. The multi-occasion half guards against false negatives. A gate resting on one thirty-minute viva amplifies task-sampling unreliability into false negatives, and those false negatives concentrate on exactly the students the architecture's equity commitments protect. The OSCE tradition in medical education — the Objective Structured Clinical Examination — exists because multi-station breadth is psychometrically mandatory. The conjunctive half takes its rationale from Section 4: the unaided foundation is the schema base that delegation judgment presupposes. A compensatory rule would let polished augmented work buy back absent foundations. The credential would then carry, inside itself, the very substitution that Section 3 made invisible outside it.

**Register B — augmented performance.** *Claim:* Professional-standard outcomes were achieved under the student's accountability, with unrestricted access to frontier systems. The

claim is deliberately impersonal, for a reason the paper has already supplied. A Register B artifact is produced unsupervised, in a machine-generable medium; by Section 3's own analysis, its quality, standing alone, certifies nothing about any person. It does not even certify the direction behind the work, since direction can itself be delegated (Appendix B.1). Register B evidence therefore acquires person-level meaning only inside an assessment bundle whose linked Register C channels attribute the decisions. *Evidence conditions*: Tasks are open and AI-saturated, judged on outcome quality against professional standards, with no authorship forensics; and they are tasks on which an undirected model produces mediocrity. *Decision rule*: Aggregation is compensatory within the register, so strength on one task can offset weakness on another.

**Register C — delegation judgment.** *Claim*: The student's decisions about delegating, verifying, and integrating machine cognition meet a stated standard, specified per activity family and per autonomy level. The construct and its instrument are the subject of Section 7.2; what matters architecturally is where the evidence comes from and how it is admitted. *Evidence conditions*: Register C evidence is collected in the same capstones that yield Register B evidence, but it is scored on the decisions rather than the outcomes, and it is certified only through channels that survive Section 3 — live defence and supervised instrumented elicitation. Documentation serves as scaffolding, never as evidence; Appendix B.5 derives this constraint rather than assuming it.

Two governance items complete the specification. The first concerns who defines the non-delegable core. It is set per programme by the programme's assessment board, with external-examiner and student membership; it is published with its rationale; and it is reviewed on a stated cycle. Each clause earns its place, because a core defined by fiat will not survive appeal, and a core defined forever will certify obsolete skills. The second item concerns how award decisions are made. They aggregate by profile, not by sum. Conjunctive gates apply on Register A. Compensatory aggregation applies within Registers B and C above licence thresholds. And reporting is dual-signal, with the augmented-outcome record and the secured-competence record stated separately.

**Cost.** Synchronous assessment at programme scale is the architecture's largest objection. The validity divide is what happens if the objection is answered only for wealthy institutions. The existence proof, however, is forty years old. Medical education runs OSCEs — twelve or more supervised stations per candidate, for entire cohorts — and the costs are documented. Brown et al. (2015) report roughly £355 per student. Poenaru et al. (1997) report about CDN\$70, where examiner time is institutional.

The arithmetic for this architecture can be sketched against that baseline. Take a Register A gate that aggregates three secured occasions. It costs roughly four to five examiner-hours per student. Three such gates across a degree therefore come to twelve to fifteen hours, plus retake and moderation load. The comparison that matters is with the marking load that distant-proxy grading already consumes while certifying nothing. Assume thirty modules across a degree, with one to two marked scripts each, and twenty to thirty minutes of marking per script. That existing load is on the order of twenty to forty hours per student. The figure is an unaudited order-of-

magnitude estimate, offered for comparability. The constraint is therefore real, but it is a reallocation constraint more than an affordability one.

Three commitments are non-negotiable. The first is examiner training, structure, and moderation, which are required to make the synchronous channel fair (Joughin, 1998). The second is accommodations as design commitments — format choice where the construct permits, extended time as of right, published criteria, and awarding-gap monitoring at every gate. The third is the funded certification floor, because a sector in which secured assessment is a premium product has not solved the proxy collapse but priced it.

**Table 1.** The three registers as interpretation/use arguments.

| Register                  | Claim certified                                                                                              | Evidence conditions                                                                                                     | Decision rule                                        | Principal threats                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| A — Unaided capacity      | Performance of the non-delegable core without machine participation                                          | Supervised production; interactive where construct-relevant; multi-occasion programmatic aggregation per gate           | Conjunctive gates at programme decision points       | Examiner unreliability and bias; access burdens; cost-driven dilution; under-sampling            |
| B — Augmented performance | Professional-standard outcomes under the student's accountability (person-level meaning only jointly with C) | AI-saturated open tasks; outcome quality vs professional standards; no authorship forensics                             | Compensatory within register                         | Task staleness; grading drift toward fluency; construct underrepresentation                      |
| C — Delegation judgment   | Delegation decisions meet a stated standard, per activity family and autonomy level                          | Live defence; supervised fault-seeded audits; firewall-governed telemetry as corroboration; dossier as scaffolding only | Conjunctive at licence threshold; compensatory above | Construct redundancy; station-sampling unreliability; surveillance creep; meta-delegation gaming |

## 7.2 Delegation judgment: construct, evidence model, decision

Register C certifies a construct, and this subsection builds it: definition, relation to neighbouring constructs, the theory that generates predictions about it, the evidence model, the decision rule, the data governance, and the conditions under which the construct fails.

**Definition.** Delegation judgment is the capacity to decide what cognitive work to delegate to a machine system, under what verification regime, and to take accountable ownership of the integrated result. Six observable components make it up. The first is *decomposition*: partitioning work into delegable and non-delegable components given the goals at hand, including the recognition that what is delegable for production may not be delegable for learning. The second is *tool calibration*: an accurate, domain-conditional model of the system's failure modes, including its governance boundaries. The third is *verification design*: checks whose power matches the failure modes they are meant to catch. The fourth is *reliance discrimination*: accepting machine output when it is right and overriding it when it is wrong, as distinct from blanket trust or blanket skepticism. The fifth is *provenance transparency*: an honest, auditable account of what was

delegated and why. The sixth is *accountable integration*: defending the assembled whole, and standing behind it, under questioning.

**Relation to neighbouring constructs.** The education-side parent is evaluative judgement (Tai et al., 2018), and its extension to generative AI explicitly covers appraising AI-use processes, including whether and when use is appropriate (Bearman et al., 2024). The residual difference is not the object of judgement but the decision the judgement feeds. Evaluative judgement is developed as a capability; delegation judgment is specified for a certification decision, with stakes, an evidence rule, and a licence. The reliance literature, for its part, owns the momentary advice-taking event (Lee & See, 2004; Schemmer et al., 2023), but not the upstream decisions that generate advice-taking situations. Its most consistent recent finding is a warning, and the warning is built into this design. Explanations move reliance wholesale rather than improving discrimination (Bansal et al., 2021). Interventions reduce overreliance without improving appropriateness (Bo, Wan, & Anderson, 2025). Other neighbours hold other pieces. Tankelevitch et al. (2024) theorize the metacognitive demands involved. Fügner et al. (2022) showed that delegation quality is measurable, and that it is deficient by default. Walton et al. (2025) documented the judgement events in situ. The construct proposed here is the certifiable assembly none of these provides — an assembly with a measurement theory and a licence decision.

**Theory, generating predictions.** On the account proposed here, delegation judgment is metacognition turned outward. It is classical monitoring-and-control, extended across the boundary of the brain, and it maintains three models at once: a model of the task, a model of oneself, and a model of the machine partner. The value of the extension is that metacognition's known properties become predictions. The first prediction is domain-specificity. Monitoring requires a task model built from object-level knowledge, so delegation judgment will not transfer far beyond the domains in which the student has foundations. Certification is therefore domain-situated, and a generic "AI skills" credential is a category error. The second prediction is the novice double disadvantage. Those least able to verify are least aware of it, and machine confidence inflation compounds the problem (Shaw & Nave, 2026). Student self-governance therefore cannot replace developmental sequencing. The third prediction is trainability under accountability. Discrimination responds to incentive-and-feedback structure — in the behavioural data, overrides of faulty output more than doubled under per-item accountability (Shaw & Nave, 2026, Study 3) — so a certification regime is formative, not merely selective. Bainbridge closes the loop from the automation side: the supervisor's competence decays without the practice automation removed, and a monitor without a task model "has been given an impossible task" (1983, p. 776).

**Evidence model.** Evidence-centered design supplies the discipline (Mislevy, Steinberg, & Almond, 2003), and Cheng et al. (2024) have applied it to human-AI writing. The evidence rule follows from the paper's own theorem (Appendix B.5): only behaviour elicited under supervised conditions counts as certification evidence, because an asynchronously submitted record of good delegation behaviour is itself a distant proxy. Three channels qualify.

The first channel is the instrumented audit, specified in Appendix A. In supervised stations, the student works with a provisioned assistant whose contributions contain seeded faults — or whose

accuracy is covertly randomized within the session, the paradigm Shaw and Nave (2026) validated at scale. The student's accept, repair, and override behaviour is scored against ground truth under signal-detection logic. That logic earns its place because it separates two things the construct needs separated. Discrimination is whether the student can tell sound output from unsound. Criterion is whether the student trusts too much or too little. This is the decomposition the reliance metrics of Schemmer et al. (2023) map onto. The second channel is the live defence of real project work. The student answers for the work in person — why this decomposition, why these checks, what the model got wrong and how it was caught — and the defence is scored against published rubric anchors by paired raters. The third channel is consented telemetry from institutionally provisioned platforms, governed by the firewall described below, and admitted as corroboration only. The restriction has a principled basis. Students curate what enters their record, so telemetry can evidence capability but never its absence.

**The decision rule: second-order entrustment.** Medical education already certifies judgment-bearing performance, and the machinery it uses is entrustment. To certify a trainee is to document the answer to the question “Can we trust this trainee to execute this EPA?” — an entrustable professional activity — and to translate the answer into one of five supervision levels (ten Cate, 2005, 2013). The published extensions of entrustment to AI treat the tool as the trustee (Gin et al., 2025), or treat tool-use proficiency as one coarse entrustable activity (Jacobs et al., 2023). The decision higher education needs runs the other way. The certified party is the delegator. The institution certifies the *student as entruster* — licensed to delegate defined activity families to machine systems at a defined autonomy level, having demonstrated the judgment that delegation at that level presupposes. Decisions are made per activity family, on programmatically aggregated evidence (van der Vleuten et al., 2012). They are conjunctive at the licence threshold, and initial certification grants a single autonomy level per family. To searches conducted in June 2026, this is the first specification of an entrustment decision certifying the delegation decisions themselves. Its nearest neighbours are the mirror-image and coarse-grained steps just cited.

**The firewall.** Instrumented assessment collects interaction data, and Section 5.5 indicts surveillance. The architecture survives its own indictment only by structural separation, and the separation is not left to goodwill: it is stated as licence conditions on institutions, five of them. The first is purpose limitation with an evidentiary bar. Audit and telemetry data are assessment evidence for declared constructs, and they are barred from misconduct proceedings. The second is student curation. Outside the bounded audit stations, the student curates all telemetry. The third is a prohibition on biometric or affective capture. The fourth is open specification. Station designs, seeding protocols, scoring rules, and data flows are published. The fifth is student co-ownership of records, with published deletion schedules.

Educational assessment AI sits in the high-risk class of the EU AI Act's Annex III. Although the provisional May 2026 agreement would defer the corresponding obligations to December 2027, the Act's requirements are the obvious voluntary floor. One consequence is stated explicitly. An institution unwilling to fund the firewall should not run the instrumentation. Live defence alone yields a leaner, slower, valid Register C, and that trade is explicitly licensed.

**Kill conditions.** A construct that cannot fail is marketing, so this one is specified together with the ways it can fail. The first failure mode is redundancy. If audit scores carry no incremental information beyond fluid intelligence, need for cognition, AI trust, and domain knowledge, the construct dissolves into its covariates. In the behavioural data, the first two of those are protective and the third is a risk amplifier (Shaw & Nave, 2026). The second failure mode is unreliability. If generalizability analysis shows station-sampling variance defeating decision-grade reliability at feasible station counts, certification fails on feasibility. The third is structural incoherence. The six components are hypothesized as a formative composite; if they dissociate so far that no composite licence is defensible, the licence must be re-specified. Two further failure modes — gate failure and non-persistence — belong to Section 7.3 and Section 8 respectively. Each of these conditions is operationalized in Section 8.

### 7.3 Graduated delegation

The registers say what is certified; this subsection says how students are brought to the point of certification. The natural design instinct is a two-stage rule — prohibit machine assistance during formation, then license it — on the folk-remembered calculator model. That instinct is not what the evidence supports. In the strongest relevant experiment, the arm that eliminated learning harm was not the no-AI arm but the arm with *designed* AI from day one (Bastani et al., 2025). The calculator meta-analysis itself found an isolated, explicitly unexplained anomaly — basic-skill harm for average fourth-graders under what its authors called sustained use — against benefits nearly everywhere else (Hembree & Dessart, 1986). The staged calculator curriculum, in other words, is folk policy built on an unexplained data point.

The sequence proposed here, *graduated delegation*, therefore braids three strands rather than stacking two stages. The first strand is formation under designed configurations. During a capability's formation window, students work with instrumented configurations — explanation-first tutors, delegation throttles — because substitution during formation is where the documented damage concentrates (Bastani et al., 2025; Lehmann et al., 2025; Soderstrom & Bjork, 2015). The second strand is delegation taught from the start. A bootstrapping constraint follows from the construct itself: delegation competence cannot form in students who have never delegated. Decomposition, verification design, and reliance discrimination are therefore taught and practised on low-stakes work throughout, and the offloading literature is taught to students explicitly rather than cited at them. The third strand is licensing at certification points, and it needs careful statement, because it is where this sequence is most easily mistaken for prohibition. What demonstrated unaided capacity gates is not daily tool access, which is unenforceable and, per the formation evidence, unnecessary. It gates the certified autonomy level of Register C. The reason is Bainbridge's (1983): verification presupposes the schema that unaided practice builds. The architecture treats that hypothesis as its own empirical warrant. If Register A capacity does not predict verification performance, the gate is unjustified tracking and must go (Section 8).

Three design commitments distinguish this sequence from prohibition with better branding, and one distinguishes it from tracking. Restrictions are scoped and temporary, with stated exit tests. The sequence is explicit in the syllabus, as a falsifiable learning theory. The licence is real:



delegation is taught, assessed, and celebrated as the professional capability it is. And gates carry entitlements — resourced remediation, same-term retakes, independent appeal, and published pass-rate monitoring by demographic group.

Against the AI Assessment Scale, the difference is precise. The AIAS stages task permissions, and its revision never removes AI scaffolding (Perkins, Roe, & Furze, 2025); graduated delegation stages student licences on demonstrated foundations. Both can operate in one programme. What the architecture protects can be named without inflation: the capacity to set one's own intellectual ends, to evaluate means — including delegated means — against them, and to stand behind the result. Philosophy already discusses the testimony-case ancestor of this capacity as epistemic autonomy (Matheson & Loughheed, 2021), and the serious AI-literacy frameworks already include critical evaluation and human-accountability commitments (UNESCO, 2024; Long & Magerko, 2020). The residual difference is the authority relation the competencies serve, not the competencies themselves.

#### **7.4 Relational reallocation, reported**

The final component governs the resource the rest of the architecture spends. If frontier systems commoditize transmission and routine feedback, the scarce institutional resource becomes accountable human attention — and what is scarce should be budgeted and reported, not left as a residual. The principle is simple to state: for every function delegated to machine systems, the recovered human hours are explicitly reallocated, and the reallocation is publicly reported. The surrounding rhetoric is ubiquitous. Yet searches conducted for this work found no institution that budgets, audits, or reports the reallocation. The gap between universal claim and nonexistent accounting is the finding.

Because any single metric will be gamed (Strathern, 1997), the instrument is a basket with an audit attached. Entitlement metrics state what each student can claim: a named advisor who has met the student, and a maximum time-to-human when a student is in difficulty. Ratio metrics cover advising load per staff FTE and supervision continuity. An externally moderated annual audit examines sampled advising relationships. All of it is reported alongside the institution's AI-deployment statement. The reporting also covers, reflexively, the institution's own apprenticeships — teaching assistantships, sessional contracts, doctoral supervision. An institution that dismantles its examiner pipeline while adopting this architecture is consuming the resource the architecture runs on.

The demand side already favours the policy. Students' chatbot help-seeking is falling while their preference for human help holds (Tyton Partners, 2025). And the relational evidence base predates the technology: it was argued at book length, on multi-campus interview data, before these systems arrived (Felten & Lambert, 2020; Gallup-Purdue, 2014).

### **8. Research Agenda and Falsifiers**

The paper has promised falsifiers throughout; this section delivers them. The studies below are stated for registration, grouped by what each result would kill, and offered as direct answers to

the field's published agenda (Corbin et al., 2025).

**What would kill Thesis 1.** For Thesis 1 to die, the spread of end-to-end machine production would have to stop. Stating that falsifier runs into an immediate obstacle: no current statistic measures the quantity in question, the simulation share  $\sigma$  — the fraction of assessed artifacts whose constitutive decisions have been transferred to the machine end-to-end. A claim about an unmeasured quantity cannot die, so the falsifier begins with a measurement commitment. The commitment is to list-experiment surveys that correct for concealment, triangulated with consented telemetry governed by the firewall of Section 7.2, and with detector figures demoted to corroboration. Once that instrumentation exists, the falsifier is a trajectory, not a level cliff. If  $\sigma$  stalled durably at or below current indicator levels, rather than continuing the observed rise, the dynamics argument would be undermined. What any given level of  $\sigma$  means evidentially is read off Appendix B.6 against stated decision stakes. No single threshold decides it.

Thesis 1 would also die if authentication started working. A process-authentication regime that could distinguish human from agentic work at scale, with low false positives and without demographic bias, would restore distant proxies, and the thesis would fall with them. Current evidence runs hard the other way (Scarfe et al., 2024; Weber-Wulff et al., 2023; Zhang et al., 2024). But platform-scale watermarking (Dathathri et al., 2024) is the programme to watch.

**What would kill Thesis 2.** Thesis 2 dies if the entry-level decline proves temporary, or not attributable to AI, or invisible in wages. Three lines of evidence bear on it, and each can kill it. The early-career employment declines could reverse, or could resolve into a cyclical explanation, as the payroll panels extend. The Danish non-concentration finding could replicate in firm-linked designs that use direct telemetry of agentic tooling. And the mechanism faces a price-margin test that hiring-plan stories cannot dodge: it predicts that relative entry-level wages in exposed occupations fall alongside entry headcount. The relocated-ladder hypothesis adds a prediction of its own. It expects verification-and-oversight content to rise in graduate postings while junior headcount is maintained, and postings data can score that prediction within two years.

**What would kill the construct.** Delegation judgment dies first if it turns out to be nothing new — if audit scores merely repackage abilities the field already measures. The design that tests this redundancy administers the audit alongside measures of fluid intelligence, need for cognition, AI trust, and domain knowledge, across two or more activity families. If audit scores add no incremental prediction of independently rated delegation performance in real project work, the construct is dead.

It dies second if its scores cannot be made reliable enough to certify on. The test here is the generalizability study specified in Appendix A.5. If decision-grade reliability turns out to require an infeasible number of stations, the certification claim fails.

It dies third if its gates cannot be justified. The question is whether Register A capacity predicts verification performance, measured as audit  $d'$ , conditional on general ability. The design is randomized where ethics permit, and uses regression-discontinuity designs at gate thresholds where ethics do not. A null result converts the licence link into unjustified tracking, and the sequence must be rebuilt.

It dies fourth if it fades. Certified delegation judgment must persist, and must transfer beyond the audit itself, at a horizon of two or more years. A construct that exists only inside its own stations fails everything above.

One safeguard travels with the whole programme. A fairness tripwire accompanies every gate from first deployment: the thresholds for awarding gaps are published, and a breach suspends award stakes pending review.

**What would kill the sequencing.** For the sequencing to die, graduated delegation would have to leave students no better off than either alternative — open access from the start, or prohibition followed by licence. Graduated delegation makes two predictions fit for registration. The first prediction is that students in a capability's formation window, working under designed configurations, will match open-access peers on assisted output while exceeding them on subsequent unaided performance and on later audit scores. This is the higher-education extension of Bastani et al. (2025), with delegation-judgment growth added as a third outcome. The second prediction is that students taught delegation on low-stakes work from the start will outperform prohibition-then-licence cohorts at certification. The missing experiment named in Section 6 belongs in this agenda: instrumented configurations tested against raw access, randomized within course, with unaided retention at distance as the outcome.

**What would settle the validity divide.** What settles the divide is watching, for five years, where supervised examination survives. The test is longitudinal, starting from a 2026 baseline and running five years. Track the share of award-bearing decisions that rest on supervised performance — or, as an alternative metric, examiner-hours per graduate — by institutional wealth tier. Two findings together would confirm the prediction. The first is divergence beyond pre-existing baseline differences. The second is material market share for automated-oral products in mass and distance provision while wealthy institutions expand human examination. The prediction fails — welcome — if certification floors generalize.

## 9. Limitations

What this paper is should be stated plainly. It is a conceptual synthesis, accompanied by a formal sketch and an instrument specification. It reports no new empirical data.

The evidence beneath it has limits worth enumerating. The strongest causal evidence cited concerns secondary mathematics (Bastani et al., 2025). The central behavioural anchor is an unpublished preprint, and its tasks are far from authentic academic work (Shaw & Nave, 2026). The labour-market evidence is contested at the causal step (Humlum & Vestergaard, 2026). And the survey base over-represents one national context, an imbalance that US sources of narrower scope partially offset.

The instrument carries its own caveat. It is specified to the point of administrability, but it is not validated. Its generalizability, standard-setting, and differential-functioning evidence do not yet exist. Until that evidence exists, audit results warrant formative and progression stakes only.

Two structural limits remain. The proposals draw on examiner labour in a sector that is actively casualizing it. Section 7.4's reporting requirement is a governance instrument aimed at

that political-economy problem; the paper does not solve it. And the perspective is structurally partial. This is a student-centred analysis with no student authors, written in one language, and leaning on Anglophone systems.

## 10. Conclusion

Agentic frontier systems change the grammar of higher education's questions about AI. When systems simulate the process of study as fluently as its products, every assessment built on unsupervised production in machine-generable media stops carrying information. That is not because students became dishonest. It is because the evidence stopped discriminating. When the same systems absorb the entry-level work through which graduates became experts, the financing mechanism that made workplaces co-educators quietly fails, and formation lands, unfunded, on universities.

The same analysis that breaks the old certification points to its successor. What survives is sufficient to certify three things: what a graduate can do unaided, what they can achieve with machines, and — the new obligation of the agentic era — how well they govern the boundary between the two. This paper's central wager is that the third can be made a real construct. It can be defined against its ancestors, measured by instruments that can fail, licensed through decisions that carry entitlements, and governed by rules that keep measurement from becoming surveillance. The falsifiers are stated.

Universities did not choose this assignment. But the work of building, on purpose and for everyone, the ladder that used to build itself is now structurally theirs.

## AI Use Statement

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## Appendix A. The Delegation Audit: Instrument Specification

This appendix specifies the instrumented-audit component of Register C precisely enough for a third party to administer, criticize, or refute it. It is a specification, not a validated instrument. The generalizability study and the standard-setting it calls for have not been run, and Section 8 registers them as the priority empirical work. The design synthesizes named components — covert accuracy randomization (Shaw & Nave, 2026), seeded-error reliance measurement in students (Pitts et al., 2025), signal-detection scoring, appropriate-reliance metrics (Schemmer et al., 2023), OSCE station logistics (Harden et al., 1975), and programmatic aggregation (van der Vleuten et al., 2012) — into a certification instrument. The assembly into a certification instrument is what is claimed as new. Sections A.1 through A.5 define the claim, the stations, the stimulus material, the scoring, and the decision rule; A.6 through A.9 address gaming, fairness, cost, and limits.

### A.1 Claim and scope

The first thing to fix is what one administration of the instrument does and does not claim to measure. One audit cycle yields evidence for three components of delegation judgment (Section 7.2): reliance discrimination, verification design, and — via the linked defence — accountable integration. The evidence applies to one activity family (e.g., literature synthesis; data analysis; code; argumentative drafting) at one autonomy level. The cycle does not measure decomposition or provenance transparency; those components are evidenced in project work and defence. A licence decision draws on multiple cycles; A.5 specifies how the cycles aggregate.

### A.2 Station design

The station is the instrument's unit of administration. A station is a supervised session of approximately 45 minutes in which the student completes an authentic task in the activity family, working with a provisioned assistant whose outputs are experimentally controlled. (Instrumented remote administration is permitted for formative use only; see A.6.) The task sits at the difficulty of the student's current programme stage. The station presents the student with a working context — sources, data, a partially completed project — and the assistant produces a stream of contributions: summaries, analyses, code fragments, draft passages, citations, claims. A known subset of these contributions contains seeded faults (A.3). The student's task is the authentic one: produce the integrated deliverable. The work proceeds under the explicit standing rule of Register C work — the student is accountable for everything submitted, whatever its origin.

Two protocol variants serve different measurement purposes. In the *judgment-first* variant, the student commits an initial answer or plan before the assistant's contribution is revealed. This enables the relative-reliance metrics of Schemmer et al. (2023), which require knowing what the student would have done alone. In the *workflow* variant, the assistant is available ad libitum, and reliance is read from accept/repair/override behaviour against ground truth, as in Shaw and Nave (2026). Stations end with a brief tool-free segment: two or three of the student's accept/reject decisions, selected by the examiner or sampled automatically, must be justified in writing or

orally without machine access. This segment is scored under Register A conditions, and it exists for a structural reason (Appendix B.5). It is what prevents the audit itself from being delegated to the student's own model in instrumented-but-remote administrations.

### A.3 Seeding protocol

The seeded faults are the instrument's stimulus material; this section specifies where they come from and how they are calibrated. Faults are drawn from a published taxonomy with four classes: factual (fabricated or misattributed claims, hallucinated citations), methodological (inappropriate test, leaking data, confounded comparison), logical (conclusion exceeding evidence, circularity, quantifier error), and omission (missing limitation, unstated assumption that fails in the station's context).

Three calibration rules govern seeding. (1) *Detectability-in-principle*. Every seeded fault must be detectable from materials available within the station by a student at the certified level. This is verified empirically by piloting on a reference cohort; faults detected by fewer than a stated floor of reference students at the target level are replaced. (2) *Ecological anchoring*. The seed bank is built from the documented error modes of current production models in that activity family, and it is refreshed each administration cycle. The refresh is necessary because model error distributions change with model versions: an audit calibrated to last year's failure modes certifies vigilance against the wrong threats. (3) *Consequentiality*. Faults are seeded at load-bearing points, where accepting them would corrupt the deliverable's conclusions. Detection then reflects verification of what matters, not proofreading.

Students are briefed, in advance and in writing, that audit stations contain controlled faults at undisclosed locations and rates; they are blind only to which contributions are faulty. This is the disclosure structure of simulated patients in OSCEs and of seeded items in proficiency testing: no deception about the situation, blindness only within it. Fault rates are not fixed at the 50 percent of the laboratory paradigm, which was an engineered worst case (Shaw & Nave, 2026). They are instead set per station between published bounds and randomized across students. The randomization makes base-rate guessing unprofitable: a student who assumes some fixed fraction of contributions is faulty gains no advantage.

### A.4 Scoring

Scoring converts station behaviour into signal-detection quantities and rubric judgments. Each assistant contribution the student acts on yields one of four outcomes. A seeded fault flagged and adequately repaired is a hit. A seeded fault accepted into the deliverable is a miss. A sound contribution accepted is a correct acceptance. A sound contribution "corrected" into error, or rejected with spurious justification, is a false alarm.

Two scores summarize discrimination and bias. Both are computed per station from the hit rate  $H$  and the false-alarm rate  $F$ : sensitivity  $d' = z(H) - z(F)$ , and criterion  $c = -[z(H) + z(F)]/2$ . In plain terms,  $d'$  indexes how sharply the student separates faulty contributions from sound ones, and  $c$  indexes the student's standing lean toward acceptance or rejection, independent of that separation.

A worked example shows the scores in operation. A station presents twelve assistant contributions, four of them seeded. A student flags and repairs three of the four faults ( $H = .75$ ) and falsely “corrects” one of the eight sound contributions ( $F = .125$ ). Then  $d' = 0.67 - (-1.15) = 1.82$ , a strong discrimination score, and  $c = -(0.67 - 1.15)/2 = +0.24$ . Under this convention, the positive  $c$  is a mild bias toward accepting assistant output. That mild bias protects sound contributions from spurious correction while  $d'$  carries the fault-detection load. The published band for  $c$  is bounded above precisely because strong positive  $c$  is the acceptance bias of surrender, only milder.

The surrender profile itself can be placed on the same scale. Untrained adults facing a covertly half-wrong assistant followed confidently wrong output on four of five engaged trials while overriding sound output on roughly seven percent — in a preprint’s engineered worst case (Shaw & Nave, 2026). Under this convention, that performance corresponds to  $d' \approx 0.6$  with  $c \approx +1.2$ : weak discrimination behind strong wholesale acceptance. The audit’s purpose is to certify distance from exactly that profile. The target failure pattern, in other words, is measured rather than hypothetical.

Two estimation rules apply. Extreme proportions are corrected before transformation, by adding 0.5 to cell counts — the standard log-linear correction. And no licence-bearing score is computed from fewer than twenty-four aggregated signal trials per family; single stations report formatively only.

In judgment-first administrations, relative AI reliance and relative self-reliance (Schemmer et al., 2023) are computed alongside. These metrics distinguish the student who is lifted by sound advice from the student who is dragged by unsound advice.

Verification *design* — were the checks the right ones for the threat model? — is scored separately by rubric against the station’s fault taxonomy. A student who ran reference-existence checks but no methodological check, in a station seeded with a confounded comparison, receives verification-design feedback regardless of luck in detection. Misses are weighted by consequence class. Finally, the tool-free justification segment is scored for whether the stated reasons for acceptance or rejection track the actual epistemic situation rather than the assistant’s confidence — the dissociation the behavioural literature warns about (Bansal et al., 2021).

### A.5 Aggregation and the licence decision

Performance assessment’s hard constraint is task-sampling variance: single stations do not yield decision-grade reliability, in this domain or any other. The specification therefore requires a generalizability study before stakes attach. Pilot administrations decompose score variance into student, station, occasion, and interaction components. The licence decision is then built on the resulting D-study — the decision-study projection of how many stations and occasions a decision of the required reliability needs. The working target, pending those data, is on the order of six stations across two occasions per activity family, revised by evidence.

Aggregation is defined, not assumed. Hit and false-alarm counts pool across stations within an activity family before transformation (pooled-count  $d'$ ), with per-station scores and per-fault-class

breakdowns reported alongside. The breakdowns matter because fault classes differ in detectability, and a class facet therefore belongs in the generalizability model.

Licence decisions then follow Section 7.2's rule. Initial certification grants a single autonomy level per activity family; level differentiation awaits station designs whose evidence varies by level — an honest gap, not a feature. Certification requires three things: (i) pooled  $d'$  meets the standard set by an Angoff-type panel using reference-cohort data; (ii) verification-design rubric scores meet standard; and (iii) the linked defence (Section 7.2) is passed. The defence is itself scored against published rubric anchors by paired raters, contributes its own facet to the generalizability study, and is specified to the same public depth as the stations.

Criterion  $c$  is reported as diagnostic feedback with a wide published tolerance, bounded above against surrender-pattern acceptance bias, rather than gated conjunctively — that is,  $c$  is not a fourth pass/fail condition alongside (i)–(iii). Section 7.2 establishes that the conservative failure is benign and that the criterion is base-rate sensitive; hard-gating  $c$  at station priors would therefore penalize a disposition the field environment may reward.

Licences are dated and conditional on capability generation. A material change in the provisioned systems' error profile triggers re-certification of the affected families, exactly as instrument recalibration does in laboratory licensure.

## A.6 Anti-gaming

Every assessment instrument invites circumvention; this section states the channels and what the design does about each. Licence-bearing audits are administered under physical supervision, so the student's own model cannot take them. Instrumented-remote administration is defeated by a second-device attack the architecture does not pretend to solve: real-time outsourcing of the discrimination task to an independent model on other hardware. That attack is why remote mode is formative only. The logic is the same, one level down, as Appendix B.5's evidence rule: evidence of delegation behaviour must itself be elicited under conditions that attribute it. The remote-administration restriction is that requirement applied to the audit itself.

Within-station meta-delegation — asking the provisioned assistant to audit its own contributions — is permitted, and it is partially effective by design, since production models catch some of their own error classes. This permission is why the tool-free justification segment and the verification-design rubric exist. The construct is the student's governance of the process, not unaided fault-finding, and a student whose entire verification strategy is “ask the model to check itself” will show it there.

Station banks are regenerated each cycle from the live seed taxonomy (generative freshness; Section 6), and fault locations and rates vary across students within a sitting. The remaining gaming channel — coaching to the audit format — is the benign one: preparing for seeded-fault detection by practising verification is the behaviour the instrument exists to induce. Teaching to this test is teaching.



## A.7 Fairness and accommodations

Fairness commitments attach to the stations and to the defence alike. Timing and format accommodations — extended time, modality choice for the justification segment, assistive technologies unrelated to the construct — apply as of right. The discrimination and criterion standards themselves are construct-relevant and do not flex. Differential item functioning is monitored per station and per fault class across demographic groups, with flagged stations reviewed and retired. Pass rates by group are published, per Section 7.3's entitlement commitments. The known biases of oral assessment (Joughin, 1998) attach to the defence component; they are mitigated by structure — published criteria, trained paired examiners, recording, and appeal.

## A.8 Cost profile

The costs divide into marginal and fixed, and the division determines who can realistically run the instrument. Stations run cohort-parallel in ordinary computer laboratories, with one invigilator per ~30 students for supervised administrations, and with logging and ground-truth scoring automated by construction (the faults are known in advance). Marginal human cost concentrates in two places: rubric scoring of verification design, at roughly 10 minutes per student per station, and the sampled defence. Marginal cost per student per audit cycle — six stations plus one defence, including double-rated rubric samples — runs roughly 2 to 2.5 staff-hours per activity family. That is real money across four families, and it is far from the single-essay comparison sometimes offered glibly. It is nonetheless below the OSCE costs medical education already absorbs per decision (Brown et al., 2015), since no simulated patients or clinical facilities are required.

The fixed costs are the honest remainder. Seed-bank construction and piloting, reference-cohort calibration, standard-setting panels, and differential-functioning monitoring are testing-organization functions. Calibration in particular requires a cold-start sequence: initial detectability floors are set on faculty and advanced-student pilots, then re-anchored as certified cohorts accumulate. The realistic delivery vehicle is a cross-institutional consortium or assessment body maintaining shared seed banks and pooled calibration data. Such a body is also where Section 6's capability stewardship and Section 7.1's certification floor become the same project.

## A.9 Limitations

The specification inherits the validation gap of its components. The covert-accuracy paradigm was validated on short reasoning items in controlled settings, and its authors flag the CRT-only task battery and the laboratory environment as limits on generalizability (Shaw & Nave, 2026, pp. 48–49). Its transfer to extended, authentic 45-minute domain tasks is therefore an assumption this design tests rather than relies on. The G-study, standard-setting, and differential-functioning evidence do not yet exist. Until they do, audit results should carry formative and progression stakes, not award stakes.

The instrument samples behaviour, not disposition. A student can perform vigilance for 45 minutes and surrender for the rest of the term. This gap is why the architecture pairs stations

with project-embedded defence, and why curated telemetry is corroboration rather than triangulation: by the Section 7.2 firewall, telemetry can corroborate capability but never detect its absence. The persistence question — does certified delegation judgment survive beyond the audit? — belongs to Section 8’s longitudinal study, on which the construct ultimately stands or falls.

## Appendix B. Formal Note: Evidential Decay of Distant Proxies

This appendix states the mixture model behind Section 3.3, derives the three results used in the text, and records the signalling-theoretic dual with explicit boundaries on what the imported economics does and does not license. The model is deliberately minimal. Its purpose is to make the collapse claim checkable and its comparative statics explicit, not to add notation. B.1 sets up the model; B.2 through B.5 state the results; B.6 supplies the adoption dynamics; B.7 records the economic dual.

### B.1 Setup

The model needs four primitives — a competence type, an artifact quality, a production channel, and an adoption rate — and two objects derived from them. This subsection defines each in turn. A student has latent competence  $\theta \in \{H, L\}$  (binary for transparency; nothing below depends on it). The grader observes an artifact of quality  $q \in \mathbb{R}$  and must draw an inference about  $\theta$ . The artifact is produced through one of two channels: *unaided* (U), in which quality is generated by the student, or *simulated* (S), in which quality is generated end-to-end by a frontier model. Assistance and co-production, Section 3.3’s intermediate categories, mix the channels within one artifact; the binary isolates the mechanism, and the conclusions extend by continuity.

Let  $\sigma_\theta \in [0, 1]$  denote the share of type- $\theta$  students submitting machine-produced work. Unaided quality has density  $f_U(q \mid \theta)$ , with the standard monotone-likelihood-ratio property:  $\ell(q) \equiv f_U(q \mid H) / f_U(q \mid L)$  is increasing in  $q$ . In words, better students tend to produce better work; this is what makes grading informative at all. Simulated quality has density  $g(q)$ , with a property that is the model’s central — and empirically grounded — premise:

**Severance (scoped).** *For the simulation channel — undirected, end-to-end machine production on tasks within the model’s competence —  $g(q)$  does not depend on  $\theta$ : the quality distribution is a property of the model and the task, not of the student who submits it.*

Severance is what distinguishes machine simulation from earlier circumvention. A ghostwriter’s output varied with price and the buyer’s diligence; a frontier model’s undirected output at a given date is, to first order, type-independent. Cosmetic post-editing perturbs  $g$  but does not restore dependence on  $\theta$ .

These primitives combine into the two objects the results below manipulate. The grader observes  $q$  drawn from the mixture

$$p(q \mid \theta) = (1 - \sigma_\theta) f_U(q \mid \theta) + \sigma_\theta g(q),$$

and the diagnosticity of the artifact is the likelihood ratio

$$\Lambda(q) = p(q \mid H) / p(q \mid L).$$

By Bayes' rule, posterior odds equal prior odds times  $\Lambda(q)$ ;  $\Lambda(q) = 1$  means the artifact moves no beliefs.  $\Lambda(q)$  is therefore the right measure of what an artifact is worth as evidence: it is the multiplier the artifact applies to whatever the grader believed before reading it.

With the mixture and the likelihood ratio defined, the scoping of the severance premise can be addressed precisely. The scoping matters, and resolving it does architectural work. Where production is *directed* — the co-production channel, in which the student's decomposition, prompting, and integration decisions shape the output — quality retains dependence on a skill. That skill is not the certified construct  $\theta$  as traditionally conceived; it is direction skill, the very competence Section 7.2 names delegation judgment. Three consequences follow. First, for the simulation channel the results below hold as stated. Second, for mixtures including co-production,  $\Lambda(q)$  converges not to 1 but to the *direction-skill likelihood ratio*: in the limit of universal machine mediation, an unsupervised artifact carries information about at most one thing — how well the production was directed. Third, even that residual information is unusable for certification at a distance, because direction itself can be delegated to a more agentic system or purchased, and an asynchronous artifact cannot show whose direction it carries. The formal core thus derives the architecture rather than contradicting it. Unsupervised artifact quality certifies no person, and Register B's claim is accordingly impersonal (Section 7.1). The surviving person-level information is direction skill. And direction skill must be evidenced through channels that attribute it — which is Register C.

## B.2 Result 1: uniform decay to zero information

The first result concerns the cleanest case, in which both types adopt at the same rate. Suppose first that adoption is type-symmetric:  $\sigma_H = \sigma_L = \sigma$ . Then

$$\Lambda(q) = [(1 - \sigma) f_U(q \mid H) + \sigma g(q)] / [(1 - \sigma) f_U(q \mid L) + \sigma g(q)].$$

For any  $q$  in the support of  $g$ ,  $\Lambda(q) \rightarrow 1$  as  $\sigma \rightarrow 1$ . In words: as machine production becomes universal, every machine-producible quality level becomes equally likely from either type of student, and the artifact stops discriminating. Moreover the convergence is monotone. Differentiating in  $\sigma$  shows that wherever  $\ell(q) > 1$  and  $g(q) > 0$ ,  $\Lambda(q)$  is strictly decreasing in  $\sigma$ , and wherever  $\ell(q) < 1$  and  $g(q) > 0$ , strictly increasing. So  $|\log \Lambda(q)|$  shrinks monotonically in  $\sigma$  at every quality level *in the support of  $g$* . Outside that support,  $\Lambda(q) = \ell(q)$  regardless of  $\sigma$  — the refinement Result 2 exploits; a full-support  $g$  recovers the unqualified claim.

Rising simulation prevalence does not merely add noise; on the machine-producible quality range it compresses the information content of the artifact toward zero. The grader's posterior converges on the prior: in the limit, reading the work teaches nothing about the student. This is the formal content of Section 3.2's principle, and it makes explicit that the collapse is a function of prevalence  $\sigma$ , not of possibility ( $\sigma > 0$ ). That distinction answers the contract-cheating objection quantitatively: pre-AI outsourcing held  $\sigma \approx 0.02$  (Section 3.3), at which  $\Lambda$  barely departs from  $\ell$ . Possibility alone never collapsed the signal; prevalence does.

### B.3 Result 2: the surviving signal is negative

The second result follows from the shape of  $g$ . Frontier models produce polished work. Formally, suppose  $g$  concentrates on high quality:  $g(q) \approx 0$  for  $q$  below some threshold  $\bar{q}$ , while for  $q \geq \bar{q}$  the simulated channel contributes mass comparable to or exceeding the unaided channels. Then the mixture behaves asymmetrically on either side of  $\bar{q}$ :

- **Below  $\bar{q}$**  (the flawed, idiosyncratic, characteristically human region): the simulated terms vanish, and  $\Lambda(q) = [(1 - \sigma_H) / (1 - \sigma_L)] \cdot \ell(q)$ . The artifact retains the full unaided likelihood ratio, rescaled by relative non-simulation rates. Visible struggle still discriminates.
- **At and above  $\bar{q}$**  (the polished region): where  $g$  dominates both numerator and denominator,  $\Lambda(q) \rightarrow \sigma_H / \sigma_L$  — a constant. Quality variation above the machine floor carries no information about competence; only the adoption ratio does. Under type-symmetric adoption the constant is exactly 1.

Hence the decay is not uniform across quality after all, once  $g$ 's shape is acknowledged; this refines Result 1's picture rather than contradicting it. The high-quality region pools first and completely, while the low-quality region remains informative longest. Excellence proves nothing; difficulty still proves something. Grading traditions that treat quality as the signal are, in the polished region, reading noise.

### B.4 Result 3: inversion under asymmetric adoption

Now let adoption differ by type. The premise here is a plausibility argument, and it is stated as such. The private return to simulation is largest where the capability gap is largest, and the experimental record shows AI effects concentrating along prior-capability lines. Assistance compresses observable gaps while in use (exploratory dispersion analyses in Bastani et al., 2025, cf. Section 5.2), and learning effects widen prior-knowledge gaps (Lehmann, Cornelius, & Sting, 2025). So consider  $\sigma_L > \sigma_H$ , holding the conclusion conditional on it.

Under that ordering the consequence is inversion, not mere decay. In the polished region,  $\Lambda(q) \approx \sigma_H / \sigma_L < 1$ : high observed quality becomes evidence *against* high unaided competence. Symmetrically, the low-region scaling factor  $(1 - \sigma_H)/(1 - \sigma_L)$  exceeds one. Two practical consequences follow. First, the grader heuristics of the enforcement era — suspicion of work that is “too good,” lenience toward visible struggle — are not paranoia but rational inference under an asymmetric mixture. Second, the victims of those heuristics are systematically the students whose *unaided* work resembles model output. That class includes the genuinely excellent and, given detector bias against low-perplexity prose (Liang et al., 2023), non-native writers whose competence is real. The inversion result thus formally connects the validity collapse to the procedural-justice harms of Section 5.5: both are projections of the same mixture.

### B.5 Result 4: what restores information

$\Lambda(q)$  collapses because the grader cannot condition on the production channel. Information is restored exactly by restoring conditioning, and the available conditioning strategies generate the architecture of Section 7. Four are on the table; the first two yield registers, and the last two are examined and set aside:

1. **Observe the channel directly.** Supervised production sets  $\sigma = 0$  by construction. This is Register A, and it is the only strategy that certifies *unaided* competence.
2. **Observe behaviour whose distribution depends on the decision locus.** Suppose a task elicits behaviour B — responses to seeded errors, verification choices under covertly randomized assistant accuracy, defence under questioning — whose distribution differs between students who made the constitutive decisions and students who did not. Then B carries the information  $q$  no longer does. This is Register C. An immediate corollary follows, derived rather than asserted: B itself must be elicited under supervised or instrumented conditions, since an asynchronously submitted record of B is itself a distant proxy and falls under Result 1. The framework's own theorem thus constrains its own evidence model.
3. **Grade the negative signal.** Result 2 implies flawed work is informative. But an assessment regime built on rewarding visible struggle is incentive-incompatible — it pays students to perform difficulty — and Goodhart dynamics (Strathern, 1997) would consume it. Noted and rejected.
4. **Restore the cost asymmetry per artifact.** Detection and watermarking attempt to make simulation costly or visible again. Section 3.3 reviews why this programme has failed in the field and is limited in principle. Formally, it amounts to driving  $\sigma$  back down by raising detection probability, and the observed equilibrium is the reverse.

### B.6 Adoption dynamics

The results so far take the prevalence  $\sigma$  as given; this subsection asks where  $\sigma$  is headed, and why. Let the net private return to simulation for a student be  $R = b - c - pD - m$ : grade-and-time benefit  $b$ , price  $c$ , detection probability times penalty  $pD$ , and norm cost  $m$ . Contract cheating's century at  $\sigma \approx 0.02$  was sustained by three of these terms. The price  $c$  was positive — a cash price per artifact, against today's near-zero marginal cost. The expected penalty  $pD$  was nonzero, because a human counterparty can be traced or can extort. And the norm cost  $m$  was high, because the line crossed was an unambiguous bright one.

Frontier systems move every term. They take  $c$  to approximately zero inside existing subscriptions. They take  $p$  to approximately zero in the field: 94 percent of injected machine submissions went undetected (Scarfe et al., 2024). And they erode  $m$  by embedding simulation at the end of a continuous gradient of permitted assistance (Section 3.3's taxonomy). Every term moves one way.

The observed trajectory is consistent with monotone migration toward the limit of Result 1. Direct-inclusion self-reports rose  $3 \rightarrow 8 \rightarrow 12$  percent across 2024–2026 (Stephenson &

Armstrong, 2026). Detector-side high-AI shares rose  $3.3 \rightarrow \sim 15$  percent (Turnitin, 2026). That is six to ten times the contract-cheating base rate within three years of capability availability.

A worked illustration makes the stakes of the trajectory concrete. Suppose at a polished quality level  $q^*$  the unaided likelihood ratio is  $\ell(q) = 4$  — *competent students are four times likelier than less competent students to produce such work unaided* — and the machine produces such quality four times more readily still,  $g(q) = 4 \cdot f_U(q|H)$ . Under type-symmetric adoption,  $\Lambda(q) = (4 + 12\sigma)/(1 + 15\sigma)$ ; the expression follows directly from B.1’s mixture with these parameter choices. At the contract-cheating base rate  $\sigma = 0.02$ ,  $\Lambda \approx 3.3$  — modest degradation. At  $\sigma = 0.12$ ,  $\Lambda \approx 1.9$ . At  $\sigma = 0.15$ ,  $\Lambda \approx 1.8$ : the polished artifact’s diagnosticity roughly halved. At  $\sigma = 0.30$ ,  $\Lambda \approx 1.4$ . At  $\sigma = 0.60$ ,  $\Lambda \approx 1.1$  — approaching worthlessness. Whether a likelihood ratio of 1.8 is ‘safe’ depends on the decision and its stakes. For a low-stakes formative judgment, perhaps. For an award-bearing single-artifact decision, it is far below any defensible evidential standard.

The model makes the falsifier precise. Thesis 1 is undermined if  $\sigma$ , measured with the Section 3.3 taxonomy under the Section 8 instrumentation, stalls at a level at which  $\Lambda$  in the polished region remains usably far from 1 for the decision class at issue. Under this parameterization, that level sits, roughly, below ten percent.

## B.7 The signalling dual, with boundaries

The validity argument has an exact economic dual. In Spence’s (1973) model, education separates types only because the signal is differentially costly (single-crossing). For artifact-producible signals, simulation compresses the cost ratio toward one — both types produce polished artifacts at near-zero cost — and separation through the artifact fails.

Kurlat and Scheuer (2021) characterize signalling equilibria when some evaluators also hold *direct* information about candidates. As the signal’s cost differential narrows or evaluator expertise improves, equilibrium signalling falls — “direct information substitutes for traditional signalling” (p. 822). For cost ratios near one, equilibria exist in which sorting is done entirely by direct expert evaluation (their Proposition 6).

Two of their further results do work in this paper. First, the all-signalling equilibrium *always* exists: an economy can remain locked in wasteful credentialing even when direct evaluation capacity is available — “an extreme form of coordination failure” (p. 802). This is why the persistence of distant-proxy credentials is not evidence of their informational value, and why coordinated institutional moves matter (Section 7). Second, at the level of the individual evaluator, signal and direct assessment are complements even though they are substitutes in aggregate (p. 820). Direct evaluation does not make credentials worthless; it disciplines what they must contain.

Boundaries are stated plainly, because the import is only honest with them. Kurlat and Scheuer’s model is pure theory in which education is *purely wasteful* signalling; it deliberately excludes the human-capital component this paper spends Sections 4–7 defending. Their experts are hiring firms, not universities or third-party certifiers. The model is static, contains no AI, and treats signal costs as exogenous. The mapping used here is this paper’s interpretive extension, not their result. It runs as follows: AI compresses  $c_H/c_L$  for distant artifacts; supervised

performance is a signal whose cost differential survives because it cannot be delegated; and universities administering such assessment act as the market's expert-evaluation capacity. What their theorems license is narrower and still valuable. In a market with any direct-evaluation capacity, signals that lose their cost differential lose their equilibrium role; the transition can fail to happen by coordination failure alone; and evaluation capacity and credentials end up complements at the point of use.

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